

Call for Expressions of Interest	
<i>Date of Issue:</i>	September 2026
<i>Expected Completion Date:</i>	May 2027
<i>Project Title:</i>	AI + AR Enhanced CICS Transaction Routing Documentation
<i>Company:</i>	IBM
Contact Details	
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Additional Info	
<i>Students must sign NDA:</i>	☐
<i>Project uses personal data:</i>	☐
<i>Custom hardware required:</i>	☐ If Applicable: Please provide details
<i>Relevant research areas (if applicable)¹:</i>	Augmented reality; mobile application development; human-computer interaction; software engineering; AI-assisted learning; enterprise systems; mainframe education; natural language interfaces; knowledge retrieval. IBM Bob download and introAny 2 IBM AI badges on page 15 of this presentationFollow onboarding process on page 17 of this presentationCICS doc - https://www.ibm.com/docs/en/cics-ts/6.x?topic=routing-overview-transaction
Project Background Information	
IBM CICS Transaction Server for z/OS is a core enterprise transaction processing environment. For new learners, CICS concepts such as regions, transaction routing, terminal-owning regions (TORs), application-owning regions (AORs), and relay transactions can be difficult to understand from static documentation alone. This project asks students to create a mobile application that brings a CICS documentation diagram to life using augmented reality and an AI question-answering experience. The recommended source diagram is Figure 1, “The elements of transaction routing”, from the IBM CICS TS 6.x documentation page “Overview of transaction routing”. The diagram is suitable because it shows a terminal connected to one CICS system running a user transaction in another connected CICS system, with communication handled by a CICS-supplied relay transaction. Students should use this diagram as the primary visual target for the AR experience and may use the deeper “Component reference: Transaction routing” page as a stretch reference.	

¹ An overview of different research areas in the school of computer science can be found [here](#).

In the final application, a user should be able to point their phone at the selected CICS documentation diagram and see animated transaction/message flows that explain how a request travels from a terminal to a terminal-owning region, through routing/relay logic, across an inter-region connection, and into an application-owning region. The app should also allow the user to tap visual hotspots and ask questions about the parts of the diagram.

Students must use IBM Bob during development of the project. IBM Bob should be used as part of the software engineering workflow, for example to help interpret project requirements, break down user stories, design technical tasks, generate prototype code where appropriate, review code, create test cases, and improve documentation. The final report should explain how IBM Bob was used and include evidence of meaningful use during the development process.

Requirements

The group must provide:

- A working mobile application prototype that recognises the selected CICS transaction-routing diagram and overlays AR enhancements.
- Animated transaction/message flows showing how a request moves between the terminal, terminal-owning region (TOR), relay transaction/routing logic, inter-region connection, and application-owning region (AOR).
- Interactive AR hotspots for the main diagram components, each providing concise explanatory content suitable for a learner new to z/OS and CICS.
- An AI assistant or question-answering feature that allows users to ask natural-language questions about the selected diagram and the CICS concepts it represents.
- A clear explanation of the chosen source diagram, including why it is suitable for AR enhancement and how the team avoided altering the underlying IBM documentation.
- Use of IBM Bob throughout the development process. The team must document how IBM Bob supported design, implementation, testing, debugging, and project documentation.
- A simple architecture/design document covering the mobile app, AR recognition method, content/question-answering approach, and any restrictions or assumptions.
- A short demonstration video showing the mobile app recognising the documentation diagram, displaying the AR overlay, animating transaction/message flows, and answering questions.
- A final technical report including installation instructions, user guide, test evidence, accessibility considerations, limitations, and recommendations for future development.
- A GitHub repository or equivalent source-code submission containing the application, setup instructions, and project documentation.

Important constraints:

- The project should be zero-cost and should use free/open-source tooling where possible.
- The application should not require access to confidential IBM systems, live CICS environments, customer data, or personal data.
- The team should use public IBM CICS TS documentation as the learning source and should not scrape or redistribute documentation content in a way that breaches usage terms.
- The AI assistant may use curated content, retrieval over approved source excerpts, or pre-prepared explanations, but its answers must be clearly bounded to the selected diagram and source material.

Skills

Highly desirable:

- Mobile app development, preferably React Native/Expo, Flutter, Swift, Kotlin, or similar.
- Augmented reality or computer-vision experience, including image target recognition, overlays, and simple animation.
- Strong software engineering practice: Git, agile task management, testing, documentation, and demo readiness.
- Willingness to learn IBM z/OS and CICS concepts from product documentation.
- Practical use of IBM Bob as an AI development assistant.

Desirable:

- Experience with AI question-answering, retrieval-augmented generation, prompt design, or knowledge-base design.
- Familiarity with enterprise systems, transaction processing, APIs, or distributed application concepts.
- UI/UX design for technical learning experiences.
- Accessibility-aware design, including readable overlays, alternative text, and clear explanations for users who cannot rely on visual animation alone.
- Interest in mainframe, IBM Z, CICS, or enterprise computing careers.

Submission / Award information	
<i>Date of Submission:</i>	September 2026
<i>Date of Pitch:</i>	October 2026
<i>Notification of award:</i>	October 2026

For further information on this proposal, the team representative should in the first instance e-mail the lead contact stated above at the e-mail address provided.